LHCb Detector Masks and Field Transport

Configurable XY surfaces and the FT_withHole variants

18 October 2026

Six original setups + six FT_withHole variants
TV, RICH1, UP, MP, FT, RICH2, PicoCal, and Muon

Detector planes and logical layers

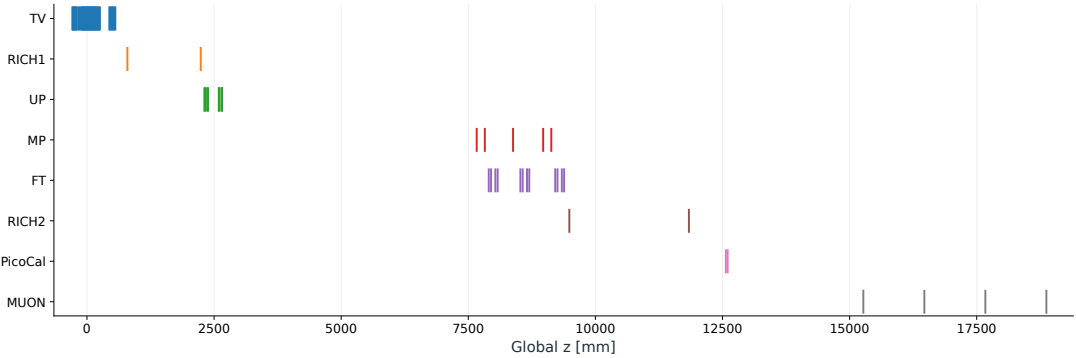
System	Surfaces	Layers	Global z range [mm]
TV	50	50	-287.5–557.6
RICH1	2	2	795.0–2237.2
UP	16	4	2314.0–2656.0
MP	5	5	7665.5–9132.5
FT	12	12	7902.7–9384.7
RICH2	2	2	9487.5–11839.2
PicoCal	2	2	12565.9–12601.0
MUON	4	4	15270.0–18870.0

Each physical surface has a global z, an XY polygon, and a stable logical layer label. A logical layer fires if any of its physical surfaces contains the transported crossing.

UP keeps 16 physical planes and counts four logical layers. FT keeps X/U/V/X in each of T1, T2, and T3: 12 logical layers.

FT_withHole keeps every plane position and label. Only the FT XY masks change.

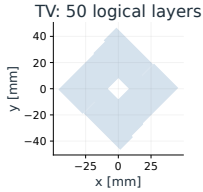
Global Z placement of the detector surfaces



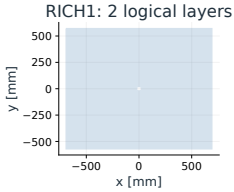
Every tick is a physical surface Z position. UP contains four groups of four surfaces; each FT station contains X/U/V/X.

The FT_withHole variant changes the XY masks while retaining all placements.

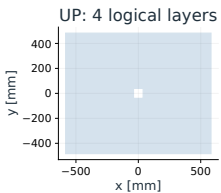
TV, RICH1, UP, and MP XY acceptance



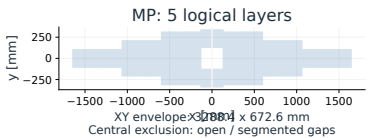
XY envelope: 90.3 x 92.0 mm
Central hole envelope: 17.5 x 17.5 mm



XY envelope: 1384.1 x 1142.5 mm
Central hole envelope: 39.8 x 39.8 mm



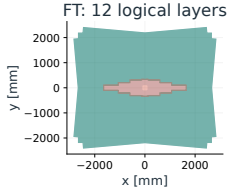
XY envelope: 1164.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm



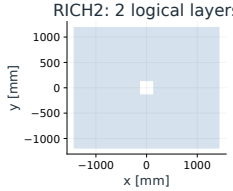
XY envelope: 3288.4 x 672.6 mm
Central exclusion: open / segmented gaps

XY unions of stored surfaces; axes are scaled independently. Open sensor gaps are shown as gaps rather than assigned a rectangular hole size.

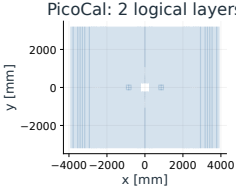
FT, RICH2, PicoCal, and Muon XY acceptance



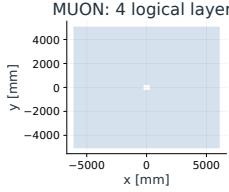
XY envelope: 5652.3 x 4798.7 mm
FT_withHole: 3288.4 x 672.6 mm



XY envelope: 2855.1 x 2375.8 mm
Central hole envelope: 284.6 x 284.6 mm



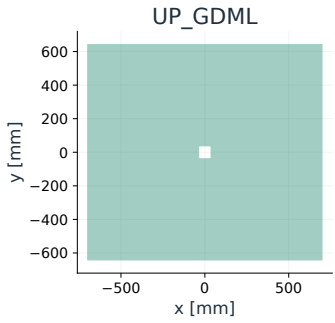
XY envelope: 7801.6 x 6338.8 mm
Central exclusion: open / segmented gaps



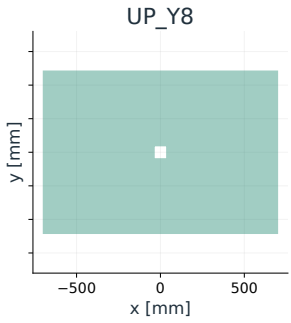
XY envelope: 12119.2 x 10057.3 mm
Central hole envelope: 608.4 x 495.1 mm

XY unions of stored surfaces. FT: nominal outline in blue, retained FT_withHole in green, stepped MP envelope in red.

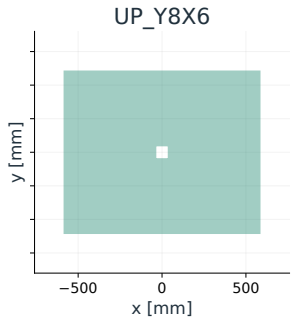
The three UP border variants



Outer: 1393.7 x 1279.8 mm
Central hole envelope: 77.3 x 79.9 mm



Outer: 1393.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm



Outer: 1164.7 x 964.9 mm
Central hole envelope: 77.3 x 79.9 mm

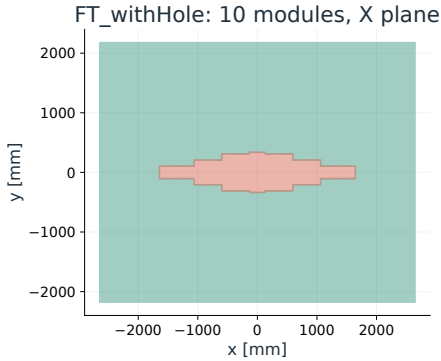
`UP_GDML` Original extracted acceptance.

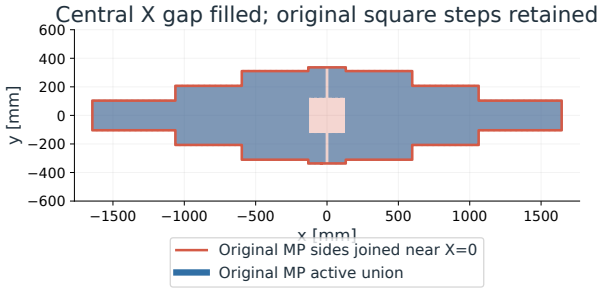
`UP_Y8` Top/bottom cutoff.

`UP_Y8X6` Top/bottom and left/right cutoffs.

FT_withHole joins the MP sides near X=0

$$H_{\mathrm{MP}} = \mathrm{join}_{x=0}\left(\bigcup_{i=1}^{5} A_{\mathrm{MP},i}\right), \qquad A_{\mathrm{FT},j}^{\mathrm{withHole}} = A_{\mathrm{FT},j} \setminus H_{\mathrm{MP}}$$



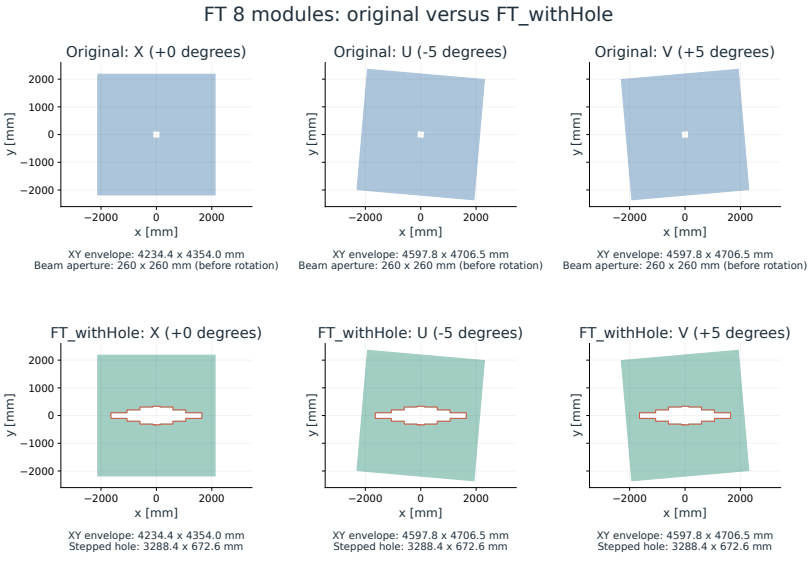


Hole envelope: 3288.4 × 672.6 mm; area 1.3806 m^2.

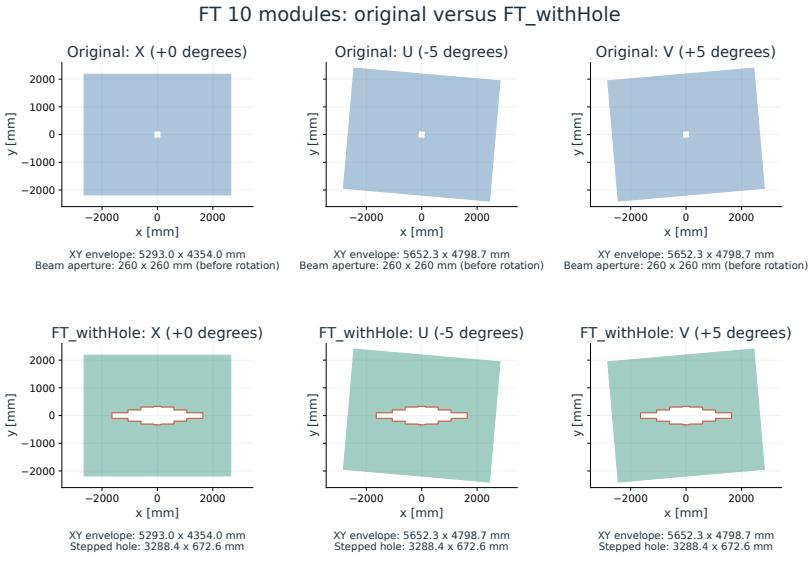
MP active union: 1.3003 m^2; the envelope additionally excludes 0.0803 m^2 across the inner gap; the outer square steps are preserved.

One global XY envelope is used at every FT plane; there is no projection in z.

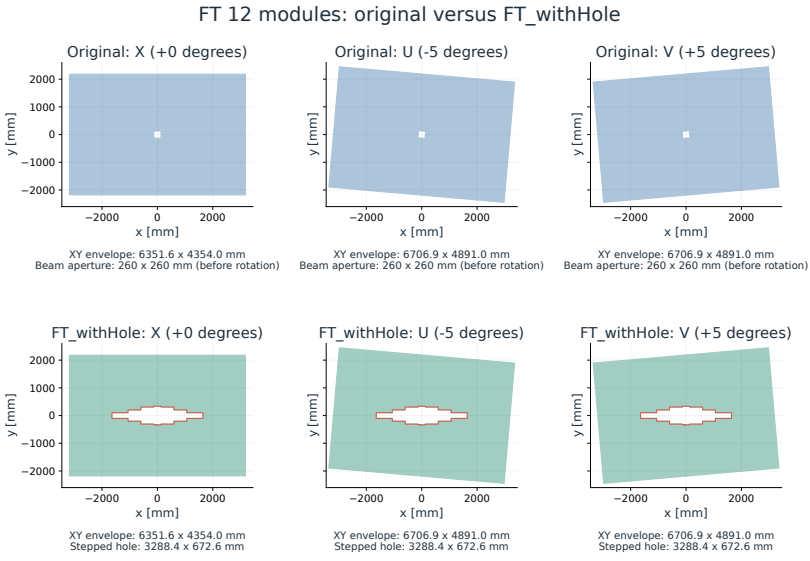
FT 8 modules: original and FT_withHole



FT 10 modules: original and FT_withHole



FT 12 modules: original and FT_withHole



Setup inventory and FT active area

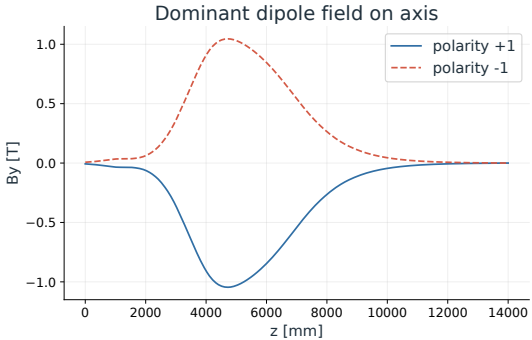
UP choice	FT modules	Original	With hole
GDML	8/8/10	Yes	Yes
Y8	8/8/10	Yes	Yes
Y8X6	8/8/10	Yes	Yes
GDML	10/10/12	Yes	Yes
Y8	10/10/12	Yes	Yes
Y8X6	10/10/12	Yes	Yes

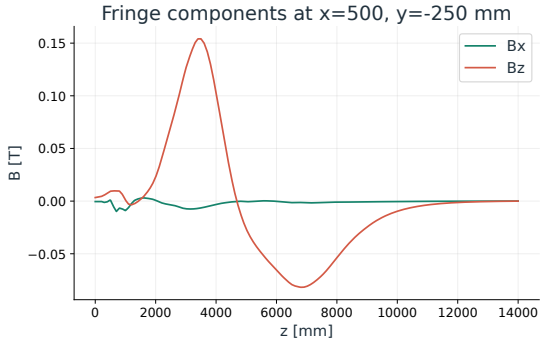
Each original catalog ID has an additional ID ending in `_FT_withHole`. Other detector masks are identical to its parent setup.

Modules	Original [m^2]	Kept [m^2]	Removed
8	18.3690	17.0560	7.15%
10	22.9781	21.6652	5.71%
12	27.5873	26.2743	4.76%

Areas are per FT plane. X/U/V rotations preserve these areas for the current MP envelope. The original area already excludes the 260 mm beam opening.

Field map and magnet polarity

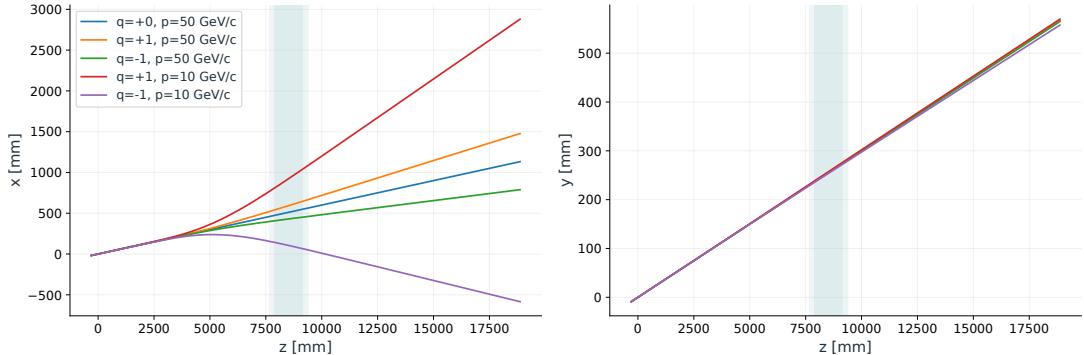




The ROOT service reads Bx, By, Bz on a 100 mm grid. Coordinates are mm, field is tesla, momentum is MeV/c.

`polarity=-1` reverses all three components, including fringe fields.
On-axis $\int_0^{10000\,\mathrm{mm}} B_y\,dz = -4.1856$ T m for the supplied map.

Toy trajectories through the full detector



Initial slopes $t_x = 0.06$, $t_y = 0.03$; start at $z = -300$ mm with the zero-field line through the origin. The same field trajectory is evaluated against both mask variants.

RK4 transport in z; cubic Hermite XY interpolation. No material effects. A nearly transverse or curling path raises `TrajectoryError`.

Fired layers before and after the enlarged hole

Reference setup: UP Y8X6, FT 8/8/10; supplied field polarity $+1$.

Track	p [GeV/c]	nMP	nFT original	nFT with hole	nUP	nMuon
Neutral control	50	5	12	0	4	4
Positive	50	3	12	7	4	4
Negative	50	5	12	0	4	4
Positive	10	0	12	12	4	4
Negative	10	5	12	0	4	4

FT_withHole can only reduce FT acceptance. MP, UP, TV, RICH, PicoCal, and Muon use the same masks and return the same flags as the parent setup.

An XY point in an MP sensor gap is also rejected by FT when it lies inside the solid envelope. Removing FT overlap does not guarantee an MP hit at a different z.

Batch and parallel propagation

Method at 10 mm steps	Per track	4 million tracks
Individual Z spline	314 ms	14.6 days
Batch of 1024, one process	1.25 ms	1.4 hours
Batch of 1024, four processes	0.45 ms	30 minutes

Measured on 18 October 2026. Batch timing: 8192 toy tracks at 10–100 GeV/c, including labels; four-process timing includes worker startup. Four-million times are extrapolations. Serial/parallel hit-count checksums agreed.

The individual Z path uses about 2005 nodes; vectorized batches use roughly 1500 main steps with longer drift steps outside the field. Reuse crossings to compare geometries sharing the same physical Z planes.

50 mm steps reduced cost further: maximum XY difference 0.0272 mm, with no changed layer flags on a 4096-track check. The default remains 10 mm.

Batch mask columns and geometry comparisons

```
columns = dict(x=x, y=y, z=z, tx=tx, ty=ty, q_over_p=qop)
batch = transport.propagate_batch(columns, label_suffix="base")
track_table.update(batch.as_columns())

# Different XY mask, same state/field/Z planes: reuse transport.
other = DetectorTransport(other_configuration, transport.field)
track_table.update(other.labels_from_crossings(
    batch.crossings, label_suffix="withHole",
))
```

A worker loads the geometry and field once, then processes chunks of 1024–4096 tracks. The process-pool example is `examples/benchmark_transport.py`.

Batch transport raises on failed/incomplete tracks, reporting their row indices.

Loading a setup and appending track flags

```
from xyshapes_detectors import DetectorTransport, TrackState

transport = DetectorTransport.from_files(
    "lhcb_lower_up_y8x6_ft_8_8_10_FT_withHole",
    "MagField/magnetic_field_v5r0.root", polarity=+1,
)
state = TrackState.from_charge_momentum(
    charge=1, momentum_mev=50_000,
    x=0, y=0, tx=0.06, ty=0.03, z=-300,
)
result = transport.propagate(state, label_suffix="hole")
track_row.update(result.as_record())
```

`result.layer_hits`: explicit logical-layer booleans.
`result.trajectory`: path and XY intercepts at requested z.
Returned fields include `nTV_hole`, `nUP_hole`, `nMP_hole`, `nFT_hole`, `nMuon_hole`, RICH flags, and `hasPicoCal_hole`.

Geometry derivation and reusable files

1. Detector surfaces and placements originate from the saved ViewLHCb/GDML acceptances and simplified FT module-count constructions.
2. `build_ft_with_hole_variants.py` joins the two MP sides near X=0 within each horizontal band, subtracts this stepped hole from each FT mask, and saves named shape pickles.
3. Every new setup stores `layout.csv`, `config.json`, and `ft_area_summary.json`. The config records the envelope vertices and operation.
4. Catalog entries reuse shared source masks and nine derived FT shapes. Both catalog IDs and setup directory paths can be loaded.

Sources: `detector_setups/catalog.json`, saved XY pickles, `variants.py`, `transport.py`, and the supplied field ROOT file.

Scope: geometric acceptance along a vacuum trajectory. This is not a material, scattering, light-transport, or detector-efficiency model.